Cluster 03 - Data-driven decision making

THE FIELD ASKS · proposed

SHARED THEMES



A PUBLIC SPACE

Elena
follows a photo of the place

Giovanni
follows a street tree in Barcelona

David
Compares native vegetation and water to improve outdoor thermal comfort.

Whose data counts?

Where does it excel? Predicting before building

Where can it fail? Partial truth, a simple rule that wins

What does it still lack?

1 WHO IS IN THE MODEL
representation - vulnerability

2 WHAT THE PLACE DOES TO THEM
comfort - impact prediction

3 HOW THE DECISION IS MADE AND CHECKED
model validation - decision support

4 WHAT IS STILL MISSING
knowledge frontier - gap type

01 THE COARSE RULE
One rule for everyone

A fixed table treats every person, or every tree, the same



- Demographic labels: age, gender, diagnosis
- A municipality table picks the species and sidewalk width; no heat or runoff

02 INSTEAD
Measure each one

The individual, not an average or a type



- Each person's own sensory account: comfort levels, what bothers them and what they like
- Each tree position scored by its own crown and leaf area (LiDAR)

03 THE PROXY
Predict before anyone is there

Through a stand-in for the real place, with its limits stated



Elena and David: "cooler" does not always mean a person feels more confortable

Giovanni and David: the same UTCI index, water as a factor

- Photo ratings: expected comfort, not comfort on site
- Simulated: a fast shade × leaf-area proxy, checked against UTCI; runoff from a 2-year design storm
- Simulate pedestrian thermal comfort using UTCI to compare vegetation and water.

04 THE GROUND TRUTH
Check against reality

Compare the prediction with what really happens



- Real participants' ratings for the same photos
- Rain interception calibrated on real trees (Anys). Heat proxy checked against Ladybug UTCI runs *updated*
- Check simulations against measurements of air temperature, humidity, wind and globe temperature.

05 THE BASELINE
Beat the simple answer

Only useful if better than the old rule or a guess



- Simple baselines: group average, chance, copying survey
- The city's real replacement mix (2018–2026) and today's trees *updated*
- Compare with existing conditions and vegetation-only interventions.

06 THE TOOL
Hand it to people

A web app for the people who decide



Giovanni and David: urban greening, inside existing constraints

- A web app for people with specific sensory needs
- A Grasshopper plug-in for designers and a web app also for non-technical users
- A QGIS, Python and microclimate simulation workflow for comparing interventions.

07 THE GAP proposed
Push the edge

Where each line leaves the state of the art, and why the gap is there

- **Empirical gap:** no test of whether a persona reflects one person's senses in public space
- **Methodological gap:** no method picks species per tree position on heat and runoff together, inside a city's replacement rules
- **David:** your gap here

Who wrote the rule, and who was left out of it?

Is a persona, or a species mean, still a type?

Can a photo show heat on the skin, or a simulation show noise?

Ground truth is partial too: one rating, one season of field data

What if the simple rule wins?

One tool for both: a route that is calm and cool?

Why does each gap exist, and who gains if it closes?

SHARED RISK · proposed

From qualitative to quantitative: what is lost?

Every line turns a quality into a number. The number decides, so what it leaves out matters.

A place felt with all the senses
→
one photo rating

A street's shade and rain
→
crown size, leaf area, one UTCI value

Walking comfort
→
one UTCI value per scenario

Who checks what the number leaves out?

CORE PAPERS

